Car.LK – Sri Lanka Car Market Analysis Dashboard

Individual Contribution Report

ST 3011 – Statistical Programming | Semester 2, AY2025/26

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This report outlines my individual analytical contributions to the Car.LK vehicle price prediction project. My work spanned four interconnected areas **dimensionality reduction (Factor Analysis of Mixed Data), cluster analysis(k-means), regression modelling and model evaluation**. Each contribution is described below with supporting visualizations. This analysis was not part of the original group project and represents a new analytical layer added to the Car.LK dashboard. (**See car price prediction page in Car.LK**)

The following models were trained on 11 features: Engine (cc), Mileage (KM), Gear (Automatic/Manual), Leasing availability, four amenity binary flags (Air Condition, Power Steering, Power Mirror, Power Window), Condition (New/Used), Fuel Type, and Province. Age was excluded due to perfect multicollinearity with Mileage (r = 1.0).

Factor Analysis of Mixed Data

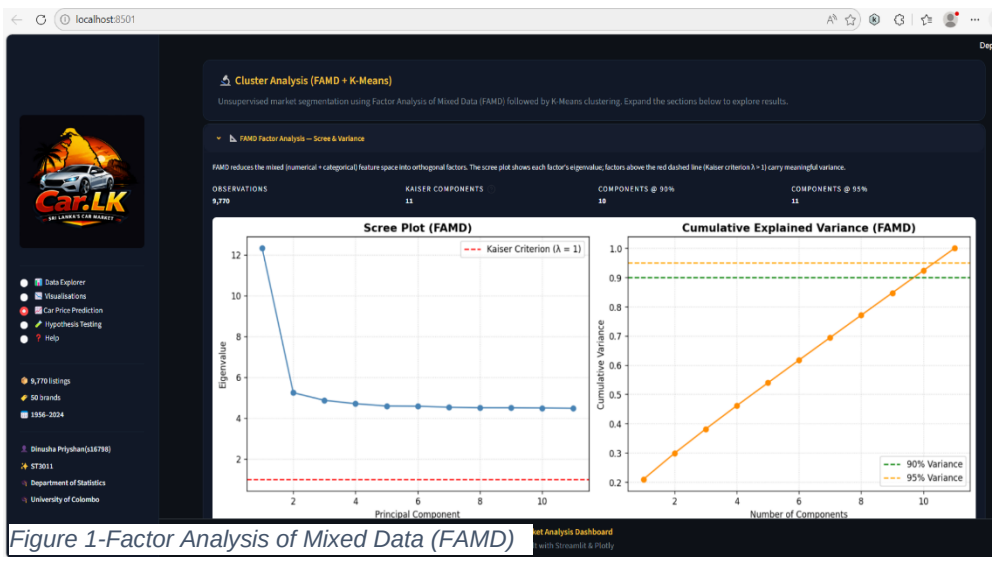


Figure 1-Factor Analysis of Mixed Data (FAMD)

The Factor Analysis of Mixed Data (FAMD) model was trained on 11 features, but capturing 90% of the variance required at least 10 factors, confirming the dataset is genuinely high-dimensional.

The first principal factor explains **20.95% of the total variance**, while the second factor explains **8.91%**, resulting in a combined variance of approximately **29.86%** for the first two components. Based on these findings, dimensionality reduction was not applied prior to predictive modelling. Instead, the **original feature set was retained** in the regression models to preserve potentially useful information for **vehicle price prediction**.

*(FAMD-python file **factor_analysis.py** and outputs saved in **factor_analysis_results.pkl**)*

K-Means Cluster Analysis

K-Means clustering was applied to the FAMD-reduced coordinates for k = 2 through k = 10. The optimal number of clusters was selected by maximizing the Average Silhouette Score.

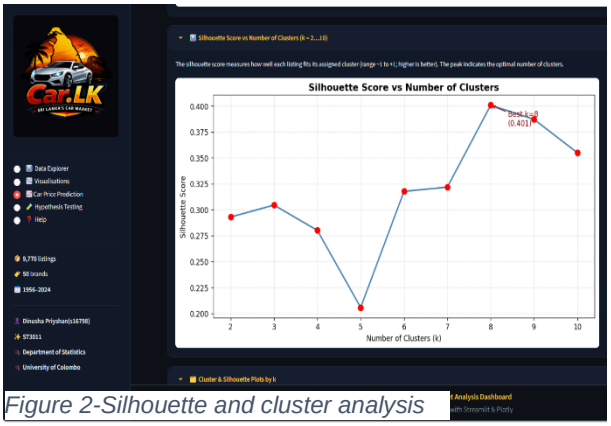
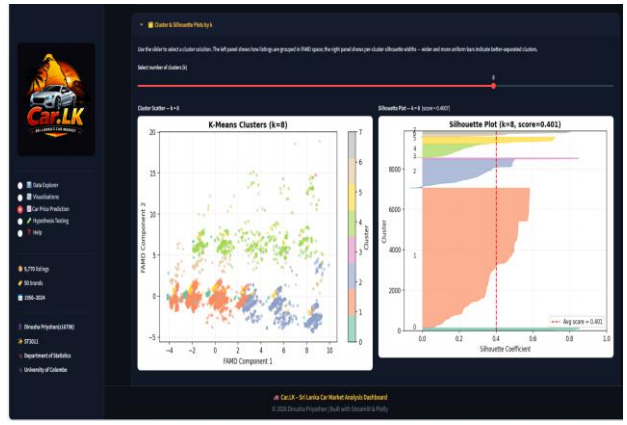


Figure 2-Silhouette and cluster analysis



Silhouette analysis identified k = 8 as the optimal cluster solution (score = 0.401), broadly corresponding to recognizable vehicle market segments from budget to luxury. However, since all silhouette scores fall in the moderate range (0.20–0.40) and the first two FAMD factors explain only ~30% of total variance, the cluster boundaries are soft and based on an incomplete representation of the data. Cluster membership was therefore not used in the regression model instead, the segmentation insights informed brand-tier feature engineering in the predictive modelling stage. For these reasons, the study focuses on **building a regression model using all vehicles rather than applying clustering-based segmentation**. *(Clusters -python file **factor_analysis.py** and outputs saved in **factor_analysis_results.pkl**)*

Regression Model Comparison & Selection

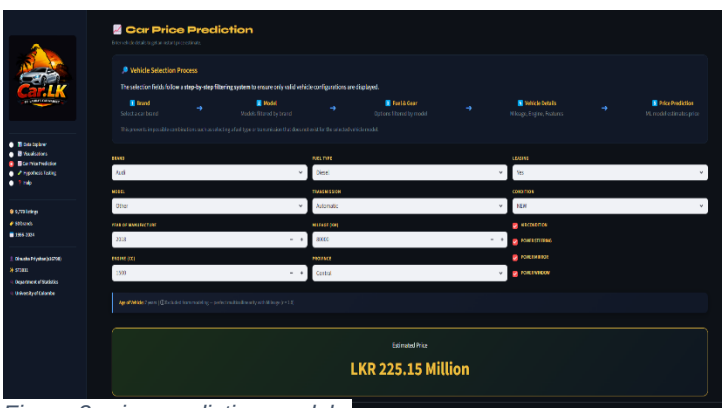


Figure 3-price prediction model

The results demonstrate a clear performance divide between linear and tree-based models. Tree-based models (Decision Tree, Random Forest, Gradient Boosting) dramatically outperform linear models, with Test R^2 of 0.64-0.68, confirming the nonlinear nature of vehicle pricing. **Random Forest was selected as the final model.** It achieves the highest Test R^2 (0.6829) and lowest Test RMSE (27.45), while showing less overfitting than Gradient Boosting (Train–Test R^2 gap of 0.197 vs. 0.217).

Model	Train R^2	Test R^2
Linear Regression	0.5237	0.4415
Ridge	0.5233	0.4403
Lasso	0.5091	.4253
Elastic Net	0.5139	0.4283
Decision Tree	0.8448	0.6442
Random Forest	0.8798	**0.6829**
Gradient Boosting	0.8901	0.6734

Table 1-Model comparison - Train R^2, Test R^2

*(Model comparison-python file **train_model.py** and outputs saved in **best_model.pkl, model_report.txt, model_diagnostics.pkl**)*

Running the Application Locally

- Run **Streamlit** app from project folder ***s16798/***, with the main file ***app.py***
- Open Terminal / Command Prompt and navigate to project folder ***cd path/to/s16798***
- Make sure Streamlit is installed, Run the Streamlit app ***streamlit run app.py***
- You will see output like: *Local URL: http://localhost:8501*
  Open the **Local URL** in your browser.